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Mapping the QTLs underlying drought stress at developmental stage of sorghum (*Sorghum bicolor* (L.) Moench) by association analysis

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Abstract This study was conducted to identify quantitative trait loci (QTLs) for drought tolerance in sorghum (*Sorghum bicolor* (L.) Moench) by association mapping using a simple sequence repeat (SSR)-marker-based diversity research set. Genotypic data for 98 SSR marker loci on ten chromosomes were used for the association analysis. The experiment was conducted under control (well-watered) and drought stress conditions, and the phenotypic values of 23 morphological traits were recorded. Drought tolerance was assessed by using a leaf drying score as a parameter of the tolerance/susceptibility: scores were assigned on a scale from 1 (most tolerant) to 9 (most susceptible). Under the control conditions, 17 QTLs associated with 12 traits were identified on chromosomes 1, 2, 4, 8, 9, and 10, with $-\mathrm{Log}_{10}(P)$ ranging from 2.5 to 7.6 and explaining 9.5–57.5 % of the total phenotypic variance for the traits. Under the drought stress conditions, nine QTLs associated with 8 traits were identified on chromosomes 1, 2, 3, and 10 that explained 9–61.2 % of the total phenotypic variance for the traits, with $-\mathrm{Log}_{10}(P)$ ranging from 2.5 to 3.5. QTLs for some traits were detected only under the drought stress condition, suggesting that these traits are important in drought tolerance. These QTLs could be used to further dissect the genetic and physiological basis of drought tolerance in sorghum.

Keywords *Sorghum bicolor* · Drought tolerance · Leaf drying score · SSR markers · Quantitative trait loci · Linkage disequilibrium analysis

Shazia Sakhi and Tariq Shehzad contributed equally to this study.

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Introduction

Drought is one of the major abiotic stresses limiting crop productivity. Worldwide water shortages and uneven distribution of rainfall make germplasm enhancement for drought tolerance important. Changes in global climate are generally considered to be occurring (Hillel and Rosenzweig 2002) and are expected to result in higher temperatures, greater evapotranspiration, and an increased incidence of drought in specific regions. An understanding of the genetic factors involved in plant responses to drought stress provides a solid foundation with which to breed plants with improved drought tolerance.

In many crops, particularly cereals, the plants are more sensitive to drought stress during the reproductive phase than at any other time, except early establishment while the root system is developing (Salter and Goode 1967). Seed set can be reduced from 35 to 75 % in various cultivars of self-pollinated wheat (*Triticum aestivum* L.) and rice (*Oryza sativa* L.) when water deficit occurs at the panicle development stage (Saini and Aspinall 1981; Sheoran and Saini 1996). Sorghum (*Sorghum bicolor* (L.) Moench) is among major cereals and has tremendous adaptation potential to abiotic stresses. It is an excellent crop for the analysis of genes contributing to environmental stress tolerance and other traits (Doggett 1988). Depending on the development stage at which it occurs, water stress has different effects on yield (Rosenow and Clark 1995; Rosenow et al. 1996; Agboma et al. 1997). Two general responses have been recognized in sorghum when subjected to drought stress, depending on whether the onset is pre- or post-flowering (Tuinstra et al. 1997; Kebede et al. 2001). A pre-flowering drought stress occurs before the anthesis stage whereas post-flowering drought stress is related to water limitation during the grain filling period. Stay-green (delayed foliar senescence) is a mechanism of tolerance to post-flowering drought stress and is characterized by the ability to avoid premature stem and leaf death (Tuinstra et al. 1997). The physiological traits affected by drought stress at the reproductive stage include flowering, peduncle elongation, panicle exsertion (PEX), and pollen fertility (POFE) (Saini and Westgate 2000; Ji et al. 2005). Drought stress at both pre- and post-flowering stages of growth has the most adverse effect on yield and yield related traits. Improvement in the performance of these traits under drought stress would help to increase grain yield in water-limited environments.

The sorghum germplasm collection is diverse and extensive, approximately 38,000 accessions maintained by International Crop Research Institute for the Semi-Arid Tropics (ICRISAT). Sorghum has a small genome size, which facilitates comparisons with the genomes of other grass species such as maize (*Zea mays* L.) and rice. For the last two decades, drought tolerance has been extensively studied in sorghum, and breeders have used 'stay-green' as a method of indirect selection for drought tolerance. Detection of quantitative trait loci (QTLs) for drought tolerance during such studies is the first step in genetic improvement to stabilize global crop production. Several studies have therefore investigated QTLs and their associated effects on drought-related traits. Most of the known QTLs for this trait were identified by using populations such as recombinant inbred lines (RILs) (Crasta et al. 1999; Kebede et al. 2001; Subudhi et al. 2000; Tao et al. 2000; Tuinstra et al. 1996, 1997; Xu et al. 2000), F_2 and F_3 populations (Habyarimana et al. 2010). Under terminal (late-season) drought, the stay-green trait has a positive impact on grain yield (Borrel et al. 2000; Jordan et al. 2003; Kassahun et al. 2009). Several stay-green QTLs are co-localized with QTLs for grain yield, flowering time, and plant height (PH) (Sabadin et al. 2012). Tuinstra et al. (1996) found six regions in the sorghum genome associated with pre-flowering drought tolerance and eight additional regions generally associated with yield or yield components under fully irrigated conditions. Four major QTLs associated with pre-flowering drought tolerance in sorghum were identified by Kebede et al. (2001). Recently, several QTLs for nodal root angle were found to be co-located with QTLs associated with drought response in sorghum (Mace et al. 2012). Drought stress at developmental stage affects panicle size, grain number, and grain yield. In sorghum, drought stress at developmental stage is common, but not enough information is available regarding the genetic control of drought stress tolerance.

QTL mapping followed by DNA marker-assisted selection (MAS) is an efficient approach for identifying genomic regions linked to complex growth traits and pyramiding the desirable alleles to improve drought tolerance in crops (Ashraf 2010). Two main approaches are used for genetic mapping: family-based linkage (FBL) mapping and linkage disequilibrium (LD)–based association mapping (Mackay and Powell 2007). FBL is a classical approach in which LD is created by developing a mapping population (e.g., F_2, doubled haploid, backcross, RILs, near-isogenic lines) which is genotyped and examined for the trait(s) of interest in different environments. In FBL the accuracy of mapping a QTL or gene depends on the size of mapping population, genetic variation covered by population and number of molecular markers applied. It has low resolution due to lower number of meiotic events happened and evaluates few alleles in a relatively longer time. In contrast, LD-based association mapping requires a collection of diverse materials that is used to associate genetic markers with phenotypes of interest, taking advantage of the lower levels of linkage disequilibrium present in linkage populations. Association mapping also takes population structure into account (Kang et al. 2008) and enables mapping of complex traits. LD-based association mapping offers a useful approach that overcomes the limitations of the FBL approach, such as the time and effort needed to develop specific populations for analysis (Ross-Ibarra et al. 2007).

Sorghum is an ideal crop for association-mapping methodologies because of its moderate level of LD and its self-pollinating mating system (Hamblin et al. 2005). A panel of diverse grain sorghum germplasm for association mapping was previously reported by Casa et al. (2008). Chromosomal regions associated with several traits such as photoperiod sensitivity, yield and yield components have been identified by association mapping using another sorghum diversity research set (Shehzad et al. 2009b; El Mannai et al. 2011). Recently, by using genome-wide association mapping, several classical loci for PH and inflorescence architecture were mapped in sorghum (Morris et al. 2012). These studies suggest that genetic improvement for drought tolerance based on existing genetic variability can be achieved by evaluating and selecting tolerant germplasm and by identifying QTLs associated with the complex traits underlying drought tolerance.

Taken together, the available results that the responses of particular phenotypic characteristics to drought stress at reproductive stage are genetically tractable traits that can be possible targets for genomics strategies, aimed at isolating underlying genes to improve drought tolerance in sorghum. Therefore, the present study was conducted with three objectives: (i) to evaluate the responses of sorghum germplasm to drought stress at developmental stage and map the chromosomal regions controlling yield-related characters, (ii) to examine variations in POFE and other yield-related traits in water stress at developmental stage among a sorghum diversity research set, and (iii) to identify QTLs associated with drought tolerance at developmental stage by genome-wide association analyses. The results of this study will contribute to the development of efficient approaches for allele mining for breeding drought tolerance in cereals.

Materials and methods

Plant materials, planting conditions, and stress treatments

In this study, we used a diverse research set of 107 sorghum accessions [Shehzad et al. 2009a; Electronic Supplementary Material (ESM) Table 1]. The experiment was conducted in a glasshouse under natural day-length conditions (12L/12D) in 2011 at agricultural research farm of the University of Tsukuba in Japan. Six seeds per accession were sown in round plastic pots (20 cm diameter $\times$ 25 cm tall) and at two-leaf stage, seedlings were thinned to a density of three per pot. All of the accessions were subjected to drought stress during boot stage of development. A randomized complete block design was used with two replicates. Plants were irrigated daily until the stress treatment began. For the drought stress treatment, water was withheld for 12 days when at least one plant per pot reached the panicle initiation stage. Normal watering was resumed when 50 % of the plants flowered or showed visual signs of wilting. The control plants received regular irrigation throughout the experiment. Soil moisture was measured daily in both the stress and control treatments by using the Hydro Sense Soil Water Content Measurement system (Campbell Scientific, Inc. 815 West 1800 North, Logan, USA).

Phenotypic assessments

Data on 23 vegetative and reproductive traits (Table 1) were recorded in this study. All traits were recorded according to sorghum descriptors (IBPGR and ICRISAT 1993 except for POFE and dry leaf percentage, which were recorded according to rice descriptors (IRRI 1996).

Leaf drying was used as a measure of sensitivity to drought, and was visually scored by calculating a leaf drying score (LDS) as follows:

LDS = (number of dry leaves/total number of leaves) $\times$ 100.

Each accession was then scored for drought tolerance/sensitivity on a scale from 0 to 9, as follows:

Table 1 Abbreviations, full names, and descriptions of the traits investigated in this study

S. no.	Abbreviations	Traits	Description
1	DTH	Day to heading	Date when 50 % of plants have begun heading
2	DEH	Delay in heading	Difference in heading between stress and non stress
3	DTF	Days to flowering	Date when 50 % of plants have begun flowering
4	DEF	Delay in flowering	Difference in heading between stress and non stress
5	TNOL	Total no. of leaf	Number of leaves on main stem
6	LL	Leaf length	Length of LL blade
7	LW	Leaf width	Width of the widest part of the LL
8	LDS	Leaf drying score	% age of total dry leaves at the end of treatment
9	CD	Culm diameter	Diameter of the middle of intern ode of main stem
10	PH	Plant height	Height of main stem at maturity
11	PEX	Panicle exsertion	Exsertion of the panicle from boot leaf at the end of flowering
13	POFE	Pollen fertility	Collection of unopened spikelet's via staining.
14	PL	Panicle length	Length from neck node to the tip of panicle at maturity
15	PWI	Panicle width	Width of the widest part of the panicle at maturity
16	NOP	Panicle no/plant	Total no. of panicle/plant at maturity
17	PW	Panicle weight/panicle/plant	Weight of the dry and clean panicle after harvesting
18	GNOP	Total grain no/panicle/plant	Total no. of clean grains after threshing
19	GWP	Total grain weight/panicle/plant	Weight of dry and clean grains after threshing
20	100GW	100 grain weight/panicle/plant	Weight of 100 cleaned and threshed seeds
21	FLL	Flag leaf length	Length from base to the tip of the leaf
22	FLWI	Flag leaf width	Width of the widest part to the leaf
23	NOT	Number of tiller/plant	Number of tiller at the base of the main stem/plant

0 = no dry leaves (not sensitive), 1 = 1–10 % (extremely low), 2 = 11–20 % (very low), 3 = 21–30 % (low), 4 = 31–40 % (slightly low), 5 = 41–50 % (intermediate), 6 = 51–60 % (slightly high), 7 = 61–70 % (high), 8 = 71–80 % (very high), 9 = 81–90 % (extremely high), or more than 90 % dry leaves with plant death symptoms. LDS was measured twice during the stress period, 6 days after the initiation of stress and the day before watering, and the latter value was used in the analyses. Table 1 lists the parameters.

Statistical analysis

Distribution analysis was performed for each trait using the JMP statistical software, ver. 9 (SAS Institute Inc. 2010). The data for the control and drought stress treatments were analyzed separately by one-way analysis of variance (ANOVA) using the PROC GLM procedure in JMP ver. 9. The Pearson correlation coefficients (r) between trait values under well-watered (control) and drought stress conditions were calculated using a procedure in SAS. The path coefficient analysis and significance level for the traits associated directly and indirectly with differences in yield components were obtained separately for the two conditions (Li 1956). The Tukey–Kramer HSD (honestly significant difference) test was performed for all traits in both environments (control and drought stressed) to test for significant differences in trait values.

Association analysis

A total of 98 SSR markers were used for the association analysis. The markers and genotyping methods were as described in Shehzad et al. (2009b). Population structure analysis was performed using the program STRUCTURE version 2.2 (Pritchard et al. 2000). Markov-chain Monte Carlo sampling was repeated 10^6 times after 10^4 cycles of a burn-in period. The posterior probability for the number of subpopulations, J, was the largest for $J = 3$, so this value was selected after two repetitions.

A **Q** matrix, whose (i,j)-th element was represented as q_{ij}, was incorporated into the association mapping models in which the effect of population structure was considered. A kinship matrix, **K**, was calculated as the allele-sharing rates of the 98 SSR markers as suggested by Zhao et al. (2007) and used in the models that included a **K** effect. LD between SSR markers was estimated by D and r, where D is the standardized disequilibrium coefficient and r represents the correlation between alleles at two loci. The statistical software TASSEL (Trait Analysis by Association, Evolution and Linkage) ver. 2.0.1 (Bradbury et al. 2007) was used to obtain P values representing the significance of LD. To identify QTLs significantly associated with drought stress, both a general linear model (GLM) and a mixed linear model (MLM) were applied for analysis in TASSEL. In GLM, two different models were used: (1) the naïve model, in which there is no control of population structure and kinship, and (2) the Q model, which is based on population structure (Yu et al. 2006). In MLM, two models were used: (1) a model based on kinship (K), and (2) a model that took into account both population structure and kinship (Q + K). The P values obtained from all models were converted into $-\text{Log10}\ (P)$.

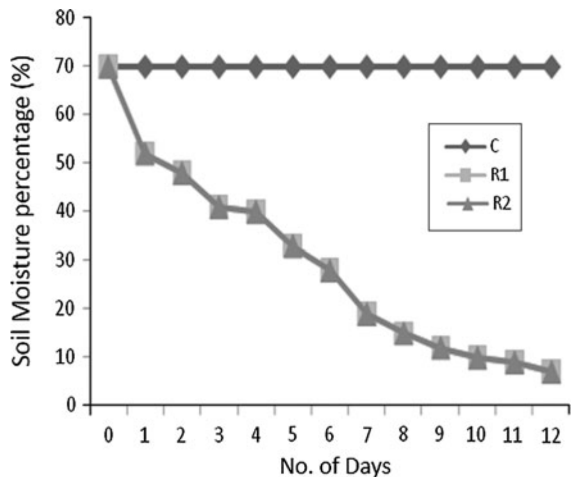


Fig. 1 Soil moisture percentage (%) measured in days 0–12 for control and treated in two replications

Marker localization and homology to known genes

SSR markers associated with traits related to drought tolerance were physically localized by BLAST in http://www.phytozome.net/sorghum, http://www.plantgdb.org/SbGDB/, or http://www.gramene.org/. Markers previously identified as linked to known genes were localized to the genome-based sequence information provided in Map Viewer at the NCBI website (http://www.ncbi.nlm.nih.gov/mapview/). The sequences with maximum matching were then used to search in Map Viewer; if the position was not given in Map Viewer, the primer sequences were then used to search the genome database in http://www.phytozome.net/sorghum. Protein sequences of genes were also used to search by BlastP, and the homologous sorghum genes were identified in http://www.plantgdb.org/SbGDB/.

Results

Moisture content

The soil moisture level in each pot of both treatments was monitored daily. Under the drought stress conditions, the average soil moisture (10–25 cm depth) decreased day by day to a minimum of 7 % compared with 60 % under the control conditions (Fig. 1). Such large decreases in soil moisture indicate that the

Table 2 Combine mean, standard deviation, and range, standard error of the mean and upper and lower confidence interval for all traits under control (CON) and stress conditions (STR)

Traits	Mean		Range (maximum–minimum)		Standard deviation		Standard error mean		Upper and lower confidence interval	
	CON	STR	CON	STR	CON	STR	CON	STR	CON	STR
DTH	60.16	59.5	105–40	115–44	13.04	31.7	1.3	3.17	62.7–57.6	65.8–53.25
DEH	–	9.75	–	36–1	–	9.21	–	0.92	–	11.5–7.92
DTF	64.31	63.19	108–44	120–47	13.63	33.14	1.36	3.31	67–61	69.7–56.6
DEF	–	10.28	–	33–1	–	9.39	–	0.939	–	12.1–8.4
CD	10.34	9.79	17.6–6.2	15–6.3	2.53	2.02	0.25	0.202	10.84–9.84	10.2–9.38
TNOL	8.95	8.27	16–6	16–4.95	2.1	1.85	0.21	0.185	9.37–8.53	8.64–7.9
LL	71.8	70.46	100–52.3	97–52	10.74	9.5	1.07	0.95	73.94–69.67	72.3–68.5
LWI	8.1	5.32	8.1–3.5	6.93–3.86	0.88	0.681	0.08	0.068	5.74–5.39	5.46–5.19
LDS	–	48.8	–	88–5	–	15.65	–	1.56	–	51.9–45.69
PH	180.2	115	353.4–53.7	216.9–45.9	53.07	35.1	5.3	3.5	190.7–169.7	122.7–108.8
PEX	2.61	1.26	1.0–4.0	3.7–1	0.682	0.851	0.06	0.085	2.75–2.47	1.43–1.09
PL	18.21	8.5	39.2–6	31.3–4.1	6.46	6.28	0.646	0.628	19.49–16.93	9.75–7.26
PWI	6	2.9	15.6–2	8.0–1	2.15	1.86	0.215	0.186	6.43–5.57	3.27–2.53
NOP	1.12	1.29	3.0–1	2.2–0	0.299	0.435	0.029	0.043	1.18–1.06	1.38–1.2
PW	11.32	2.62	41.5–1.13	27–1.1	8.17	3.67	0.817	0.367	12.95–9.7	3.35–1.89
POFE	83.02	27	99–48	89–0	12.95	26.4	1.29	2.64	85.59–80.44	32.2–21.7
GWP	9.74	2.69	25.08–2.3	18.7–0	5.18	3.47	0.518	0.347	10.77–8.71	3.38–2
GNOP	418.5	79.5	1,160–377.7	613.4–0	273.3	118.6	27.33	11.8	472.7–364.2	103.1–56
NOT	0.667	1.36	9.7–0	5.2–0	1.43	1.31	0.143	0.131	0.952–0.38	1.62–1.1
FLL	37.65	36.1	59.7–16	60.6–14.6	9.89	10.4	0.989	1.04	39.6–35.68	38.2–34.07
FLWI	5.13	4.64	7.5–2	6.8–2	0.864	0.793	0.086	0.079	5.3–4.95	4.8–4.48
100GW	2.77	2.74	11.4–0.992	24.3–0	1.43	3.4	0.143	0.34	3.05–2.49	3.4–2.06

sorghum accessions were under water stress for 12 days in drought conditions.

Assessment of drought tolerance and variation among accessions

Leaf drying score (LDS), defined as the percentage of fully dry leaves on each plant, was used as a parameter for the assessment of drought tolerance/sensitivity (see Materials and methods). Under the drought stress conditions, LDS ranged from 5 to 88 % (average of 49 %) while in the control treatment, no dry leaves were observed (Table 2). The level of variation in drought tolerance is reflected by the shifts in distributions for many of the other measured traits (Fig. 2). On the basis of the tolerance/sensitivity scale, one accession from Asia (No. 66) showed extremely low symptoms of sensitivity and less response to drought stress; at the other extreme, two accessions from Asia (No. 81) and Africa (No. 84) showed extreme sensitivity to drought stress (ESM Fig. 1).

An analysis of variance (ANOVA) showed wide range of variation among accessions for many traits under drought stress (Table 3) than under control conditions (data not shown). The results show highly significant variation for many traits under the drought stress conditions than under the control conditions. All traits were also tested by Tukey's HSD test, which gave almost the same result (Table 3).

Variation in flowering traits

Variations were observed in flowering traits between control and drought stress conditions (Table 2). Under drought stress conditions, days to heading (DTH) ranged from 44 to 115 days after sowing (DAS),

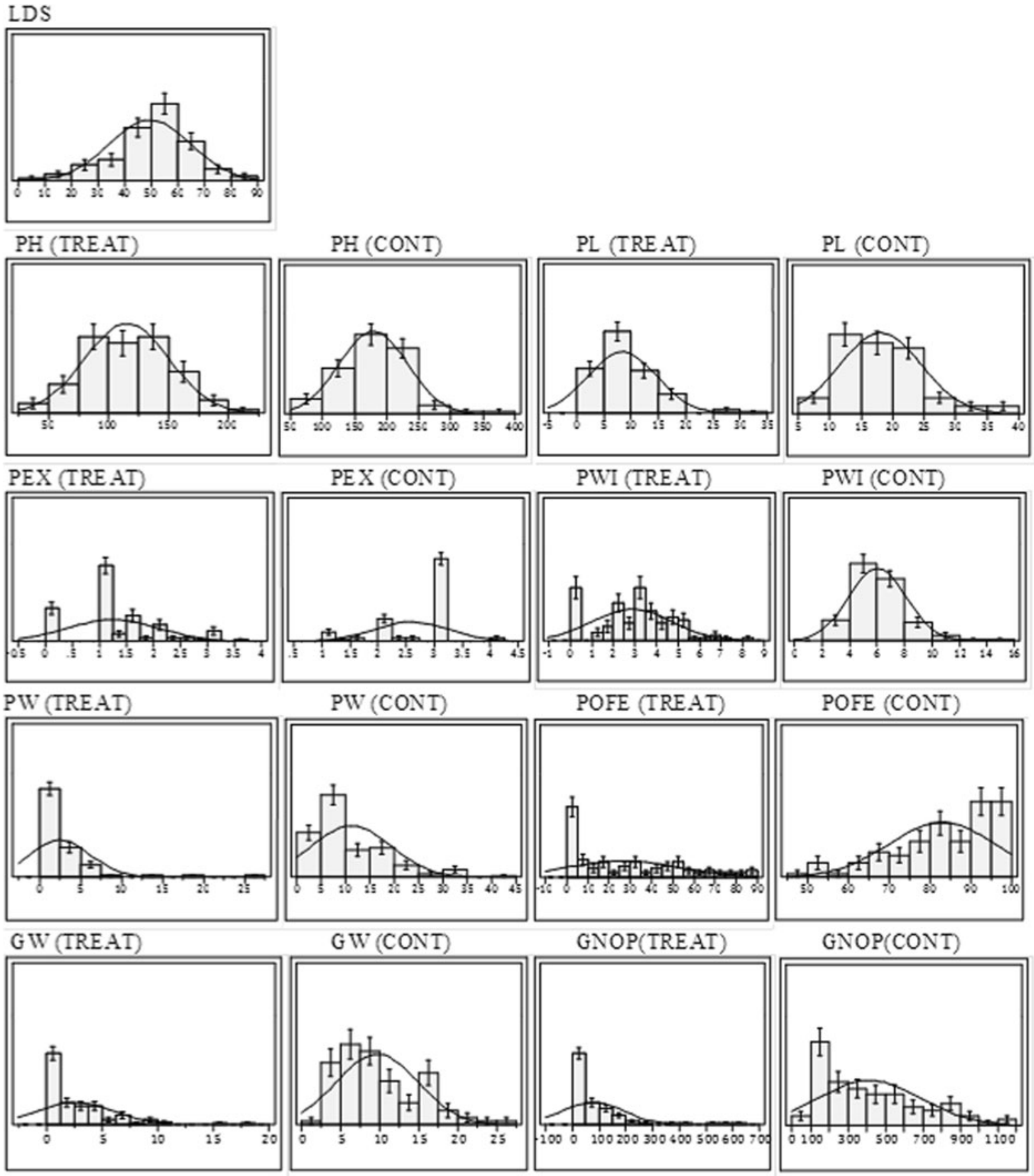


Fig. 2 Normal distribution for traits under control and drought stressed conditions

respectively, whereas the range under control conditions was 40–105 DAS (Table 2). Similarly, days to flowering (DTF) ranged from 47 to 120 DAS, whereas the range under the control conditions was 44–108 DAS (Table 2). Under the drought stress conditions, delays in heading (DEH) and flowering (DEF) were also observed in most of the accessions; values ranged from 1 to 36 days (DEH) and 1 to 33 days (DEF).

Table 3 ANOVA of the traits under drought stress conditions

Sources	*Df*	Sum of squares	Mean square	*F* ratio	$P > F$ Tukey–Kramer HSD $\alpha = 0.05$
DTH	99	424.56	106.14	0.01017	0.982[NS,a]
Error	95	99,148.72	1,043.67		
DH	99	236.96	59.24	0.6889	0.601[NS,a]
Error	95	8,169.78	85.99		
DTF	99	861.89	215.47	0.1897	0.943[NS,a]
Error	95	107,888	1,135.66		
DF	99	480.61	120.15	1.38	0.246[NS,a]
Error	95	8,259.56	86.93		
CD	99	58.92	14.73	4.02	0.005*[,a,b]
Error	95	348.57	3.67		
TNOL	99	41.61	10.41	3.31	0.014*[a,b]
Error	95	299.07	3.15		
LL	99	548.84	137.21	1.55	0.194[NS,a]
Error	95	8,418.33	88.61		
LWI	99	2.923	0.7307	1.61	0.178[NS,a]
Error	95	43.11	0.454		
LD (%)	99	2,610.22	652.55	2.86	0.027*[,a,b]
Error	95	21,639.02	227.79		
PH	99	1,824.45	4,562.11	4.18	0.004**[,a,b]
Error	95	103,831.96	1,092.97		
PEX	99	5.25	1.31	1.88	0.122[NS,a]
Error	95	66.53	0.70		
PL	99	625.87	156.47	4.52	0.002**[,a,b]
Error	95	3,286.93	34.69		
PWI	99	50.68	12.67	4.19	0.004**[,c]
Error	95	293.66	3.09		
NOP	99	3.095	0.77	4.68	0.002*[,a,b]
Error	95	15.69	0.17		
PW	99	80.22	20.05	1.52	0.203[NS,a]
Error	95	1,254.36	13.20		
POFE	99	7,375.92	1,843.98	2.82	0.029*[,a]
Error	95	62,122.08	653.92		
GWP	99	86.79	21.79	1.86	0.125[NS,a]
Error	95	1,110.33	11.69		
GNOP	99	194,162.7	48,540.7	3.85	0.006**[,a,b]
Error	95	1,199,654.9	12,627.9		
FLL	99	715.29	178.82	1.79	0.157[NS,a]
Error	95	10,019.19	105.47		
FLWI	99	1.19	0.39	0.466	0.760[NS,a]
Error	95	61.09	0.64		
NOT	99	15.33	3.83	2.36	0.059[NS,a]
Error	95	154.81	1.63		
100GW	99	24.21	6.05	0.51	0.727[NS,a]
Error	95	1,123.02	11.82		

Significantly different at ** $P \leq 0.01$ and * $P \leq 0.05$, respectively

[a] Not significantly different

[b] Significantly different

[c] Highly significant by Tukey–Kramer at $\alpha = 0.05$

A wide range of variation was observed in POFE, which ranged from 0 to 89 % (mean of 27 %) under drought stress, while under the control conditions the average was 83 %. Variation in POFE under the drought stress conditions indicates that a water deficit at developmental stage increased pollen sterility. The accessions with a lower LDS also had higher values for POFE and higher grain yield.

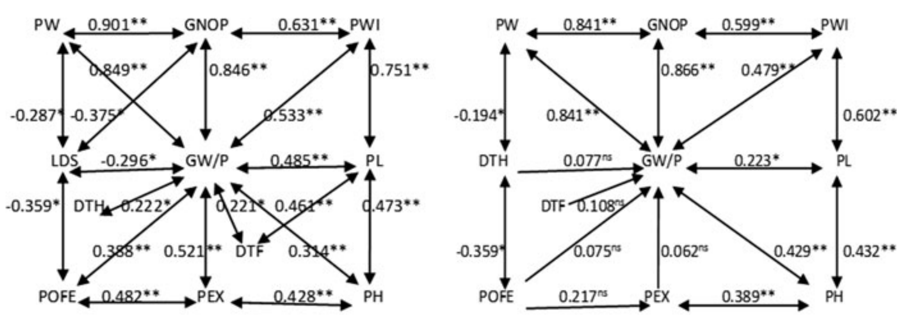
Variation in agronomic and other related traits

The mean values for the traits such as PEX, panicle length (PL), panicle width (PWI), panicle weight (PW), and grain weight per plant (GWP) showed significant reduction under the drought stress conditions (Table 2). Under the control conditions, the values for all of these traits except PEX showed a normal distribution, illustrating the variability among accessions (Fig. 2). The average grain number per plant (GNOP) varied significantly under drought stress, ranging from 613.4 to 0 with an average of 80, which showed about 81 % decrease as compared to the control condition. Among the 107 accessions measured for GNOP, only 60 accessions produced seeds, while 18 accessions did not flower, and the rest flowered but produced no seeds under drought stress. Four accessions (No. 7, 66, 23, and 65) from across Asia and one accession (No. 70) from Africa had significantly higher grain yield, which indicates tolerance to drought at developmental stage, while two accessions (No. 81 and 84) from Asia and Africa had very high LDS scores and no grain yield, which indicates extremely high sensitivity to drought. Under the control conditions, the performance of all accessions followed normal distribution for all traits (Fig. 2).

The mean PH of the accessions was lower under drought stress (115 cm) than under the control treatment (180 cm). Of the traits, i.e., culm diameter (CD), total number of leaves (TNOL), length of longest leaf (LL), and leaf width (LWI) did not show greater variation under drought conditions that may suggest that different genes are active in drought stress resistance during different development stages.

Correlation and path coefficient analysis

The phenotypic correlation coefficient (r) was estimated independently for all traits under the control and drought stress conditions (data not shown). Most of the traits were significantly and positively correlated within each of the two conditions. Path coefficient analysis was used to examine the effects of yield components such as GWP, GNOP and PW etc. with other related traits across accessions separately under each of the two conditions (Fig. 3). All the related traits contributed significantly and had a higher positive direct effect on GWP under drought stress than under control conditions (Fig. 3a). Under the drought stress conditions, LDS had strong significant negative effects on yield traits. Under the control conditions, DTH, DTF, POFE, and PEX had no significant direct effect on GWP (Fig. 3b), indicating the sensitivity of these traits to drought stress. PH had a significant positive correlation with GWP under both control and drought-stressed conditions (0.402 and



a - Traits under drought stressed conditions

b - Traits under controlled conditions

Fig. 3 Path coefficient diagram showing the effect of POFE, PEX, PL, PWI, Grain weight/panicle (GW/P), Grain number/panicle (GNO/P), DTH, DTF, PH and LDS on PW under control and drought stressed conditions. The *single headed arrow* indicates the path coefficient and the *double headed arrows* indicate the linear correlation coefficient. * and ** indicate significance at $p \leq 0.05$ and $p \leq 0.01$ respectively

0.314, respectively), which indicates that tall plants apparently had greater capacity for grain yield than short plants.

Population structure (Q)

Among the possible values for number of subpopulations (J), the posterior probability of $J = 3$ was the largest. Thus, we chose $J = 3$ and obtained estimates for the proportion of accession i's genome that originated from population j, q_{ij}. In this study, the first subpopulation ($J = 1$) was composed of 32 accessions: 27 from Africa and five from Asia. The second subpopulation ($J = 2$) was a mixed cluster that included 38 sorghum accessions of African and Asian origin. The third subpopulation ($J = 3$) comprised 37 accessions selected from East and South Asian countries.

QTL detection by association analysis

By using models in GLM and MLM, we were able to detect a large number of significant markers for traits under the control and drought stress conditions. The minimum $-\text{Log}_{10}(P)$ threshold value for a significant locus was set at 2.5 for all models under both conditions.

Detection of QTLs by GLM under control and drought stress conditions

In GLM we used two models, the naïve model and the Q model. The naïve model has no control on false positives (type I error), which can be caused by population structure, or false negatives (type II error), which can be caused by familial relatedness, so it detects a larger number of QTLs than other models. Under the control conditions, a total of 80 significant SSR loci were identified with $-\text{Log}_{10}(P)$ ranging from 2.5 to 10.6 and associated with 17 different traits. The Q model, which better controls for type I error, resolved only 17 SSR loci with $-\text{Log}_{10}(P)$ ranging from 2.5 to 7.6 and strongly associated ($P \leq 0.001$) with 12 different traits; these QTLs explained 9.5–57.5 % of the phenotypic variation for those traits (Table 4). Marker *Xtxp32* on chromosome 1 (chr-1) was associated with DTH and DTF. *Xtxp58* on chr-1 was associated with TNOL and PH, whereas three loci (*Xtxp340* on chr-1, *Xtxp315* on chr-2, and *Xtxp20* on chr-10) were strongly associated with PL. Under control conditions *Xtxp8*on chr-2 was associated with NOP explained highest phenotypic variation (57.5 %) as shown in (Table 4).

Under the drought stress conditions, the naïve model identified 63 significant SSR loci with $-\text{Log}_{10}(P)$ ranging from 2.5 to 9.7 and associated with 12 different traits. In contrast, the Q model identified only 8 significant SSR loci associated with seven traits; these loci explained 10.8–61.2 % of the phenotypic variation for those traits, with $-\text{Log}_{10}(P)$ ranging from 2.5 to 3.5 (Table 4). The Q model identified *Xtxp335* and *Xtxp149* on chr-1, *Xtxp228* on chr-3 and *Xtxp32* on chr-1were strongly associated with CD, LDS, and FLL. Two loci *Xtxp279* on chr-1 and *PepC* on chr-10 were detected for PEX with greatest phenotypic variation (61.2 %). These traits were identified only under drought stress conditions (Table 4).

Detection of QTLs by MLM under control and drought stress conditions

In MLM we used two models: the K model, which takes into consideration the relatedness between each genotype–phenotype pair in a sample, and the Q + K model, which consider both population structure and relatedness. Under the control conditions, the K model detected 34 significant loci associated with 17 different traits at $P \leq 0.001$, with $-\text{Log}_{10}(P)$ ranging from 2.5 to 19.4, whereas the Q + K model detected 17 significant loci associated with 12 different traits at $P \leq 0.001$ with $-\text{Log}_{10}(P)$ ranging from 2.5 to 7.6 and explaining 9.5–57.5 % of the phenotypic variation for those traits (Table 4). Under control conditions Q + K model detect the same loci as detected by the Q model except *Xtxp258* on chr-9 associated with LWI detected only by Q + K model.

Under the drought stress conditions, the K model resolved a large number of significant loci, second only to the naïve model, and the Q + K model detected almost the same loci that were detected by the Q model. The K model identified a total of 22 significant loci associated with 12 different traits at $P \leq 0.001$ with $-\text{Log}_{10}(P)$ ranging from 2.5 to 7.6, whereas the Q + K model identified nine significant loci associated with 8 different traits at $P \leq 0.001$ with $-\text{Log}_{10}(P)$ ranging from 2.5 to 3.5 and explaining 9–61.2 % of the phenotypic variation for

Table 4 Significant SSR loci associated with different traits under control and drought stress conditions identified by single-QTL (Q, Q + K) models of association analysis

QTLs under control conditions							QTLs under drought stress						
Trait	Model	Chromosome[d]	Location[a]	Marker[b]	$-\text{Log}_{10}$ $(P)^c$	R^2 %	Trait	Model	Chromosome[d]	Location[a]	Marker[b]	$-\text{Log}_{10}$ $(P)^c$	R^2 %
DTH	Q	1	107.1	**Xtxp32**	3.2	18.5	CD	Q	1	86.7	**Xtxp335**	2.8	10.8
		2	119.2	Xtxp1	2.5	24.0		Q + K	1	86.7	**Xtxp335**	3	9.0
	Q + K	1	107.1	**Xtxp32**	3.2	18.7							
		2	119.2	Xtxp1	2.5	21.6	TNOL	Q	3	154	**Xtxp228**	3.2	26.8
								Q + K	3	154	**Xtxp228**	3.2	26.8
DTF	Q	1	107.1	**Xtxp32**	3.3	18.7							
	Q + K	1	107.1	**Xtxp32**	3.3	18.7	LDS	Q	1	119.8	**Xtxp149**	2.5	20.0
								Q + K	1	119.8	**Xtxp149**	2.5	18.6
TNOL	Q	1	68.7	**Xtxp58**	2.6	13.0							
							PEX	Q	1	59	**Xtxp279**	2.7	18.3
LL	Q	1	53.1	**Xtxp229**	2.7	12.5			10	55.5	**PepC**	3.5	**61.2**
		4	113.2	**Xtxp51**	2.6	9.5		Q + K	1	59	**Xtxp279**	2.7	18.7
	Q + K	1	53.1	**Xtxp229**	2.7	12.4			10	55.5	**PepC**	3.5	**61.2**
		4	113.2	**Xtxp51**	2.6	9.5							
LW	Q + K	9	93.4	Xtxp258	2.7	10.8	PL	Q	2	148.7	**Xtxp315**	3.4	22.5
								Q + K	2	148.7	**Xtxp315**	3.4	22.5
PH	Q	1	68.7	**Xtxp58**	3	20.0							
	Q + K	1	68.7	**Xtxp58**	3	20.2	NOP	Q + K	10	52.5	**Xtxp270**	2.5	9.6
PL	Q	1	22.5	**Xtxp340**	3.7	21.0	FLL	Q	1	107.1	**Xtxp32**	2.8	19.0
		2	148.7	**Xtxp315**	3.4	23.7		Q + K	1	107.1	**Xtxp32**	2.6	12.0
		10	51.5	**Xtxp20**	2.8	22.4							
	Q + K	1	22.5	**Xtxp340**	3.7	21.0	NOT	Q	2	26	**Xtxp25**	2.6	11.9
		2	148.7	**Xtxp315**	3.5	23.7		Q + K	2	26	**Xtxp25**	2.6	11.8
		10	51.5	**Xtxp20**	2.8	22.4							
PWI	Q	2	122.9	**Xtxp56**	2.7	18.2							
		2	132.4	Xtxp286	2.6	14.0							
	Q + K	2	84	Xtxp19	2.6	11.0							
		2	122.9	**Xtxp56**	2.5	14.0							
NOP	Q	2	196.8	**Xtxp8**	7.6	**57.5**							
	Q + K	2	196.8	**Xtxp8**	7.6	**57.5**							
POFE	Q	4	113.2	**Xtxp51**	3.1	16.3							

Table 4 continued

QTLs under control conditions							QTLs under drought stress						
Trait	Model	Chromosome[d]	Location[a]	Marker[b]	$-\mathrm{Log}_{10}(P)^c$	R^2 %	Trait	Model	Chromosome[d]	Location[a]	Marker[b]	$-\mathrm{Log}_{10}(P)^c$	R^2 %
	Q + K	4	113.2	**Xtxp51**	3.1	16.0							
NOT	Q	2	196.8	**Xtxp8**	2.7	28.0							
		4	133.5	**Xtxp27**	2.7	22.3							
	Q + K	2	196.8	**Xtxp8**	2.7	17.0							
		4	133.5	**Xtxp27**	2.7	16.0							
100GW	Q	8	98.3	**Xtxp321**	2.9	22.6							
	Q + K	8	98.3	**Xtxp321**	2.9	22.6							

[a] Location of SSRs on linkage map as describe in (Bhattramakki et al. 2000; Kong et al. 2000; Taramino et al. 1997)

[b] Markers with bold face shows significant association with traits as resolved by Q and Q + K models

[c] $-\mathrm{Log}_{10}$ P values determined for Q and Q + K models with 2.5 as threshold value for significant association

[d] Chromosome numbers according to (Kim et al. 2005)

those traits (Table 4). For most of the traits, the range of $-\mathrm{Log}_{10}$ (P) were less than those detected under control conditions. The results of Q + K model under drought stress condition were the same as the results of Q model except for one locus *Xtxp270* on chr-10 associated with NOP.

Comparison between GLM and MLM

One of the major concerns when performing association mapping is the statistical power of the association testing and the accuracy of the identified associations because of possible false-positive and false-negative results. To take into consideration these types of errors and to check the accuracy of the QTLs mapped by GLM and MLM, we evaluated and compared these models. To evaluate each model, we plotted the observed P values (x) against the expected P values (y), as described by Stich et al. (2008). The naïve model detected the maximum number of significant loci in both the control and drought stress conditions but showed deviation from $y = x$, indicating that the results may include type I and type II errors. The K model contained the next-highest numbers of SSR loci under both conditions but still showed deviation from $y = x$. The results showed that the Q and Q + K models were more consistent in their predictions for all traits under both environments. In comparison with the naïve and K models of association analysis, the Q and Q + K models detected fewer loci, but the two models (Q and Q + K) detected almost the same loci, showing that these models had strong control over statistical errors. A model taking into account both population structure and kinship (Q + K model) appeared to be the most powerful model for population mapping studies in maize (Buckler et al. 2009). However, in the study of Thornsberry et al. (2001) the Q model was preferred in comparison with K model. In the present study, the Q and Q + K models detected almost the same QTLs within each of the two conditions. Because of this result and the statistical accuracy of the Q and Q + K models revealed in other studies, these models were used here to predict QTLs (Table 4).

Physical co-localization with known genes

To validate our results, we physically localized our markers and compared their positions with those of previously mapped QTLs in sorghum. Tuinstra et al. (1997) identified QTLs for stay-green in linkage groups (LGs) F, B and G (chr-1, 2, and 3) and yield stability in LGs B and G (chr-2 and 3) under the drought stress conditions; this finding is mostly in accordance with our findings on LGs A, B, and C (chr-1, 2 and 3) suggesting a positive relationship between LDS, stay-green, and yield traits under drought stressed conditions. One QTL (*SbAGB03*) identified in this study was physically localized on chr-2 at 58,128,106 bp and was homologous to a protein-coding gene SB02g024110 with a molecular function of binding DNA or protein. Similarly *Xtxp8* (LG B), located on chr-2 at 64,824,875 bp, was found to be within the sequence of the gene Sb02g029730, which plays an important role in ATP binding and protein tyrosine kinase activity. Further experimentation is needed to establish whether either of these loci is related to drought tolerance in sorghum.

Discussion

Abiotic stresses are major constraints to the growth and productivity of crops. The greatest abiotic stress that limits crop growth worldwide is water availability (Araus et al. 2002). The impact of drought stress on crop plants can be partly mitigated through genetic improvement. Even though the world collections of sorghum contain more than 38,000 accessions, the genetic base currently used in breeding programs is very small (about 3 % of the total world collection). Genetic improvement for drought tolerance in sorghum will require selection for tolerant germplasm and deeper understanding of the physiological and genetic responses to stress.

Performance of sorghum germplasm under drought stress conditions

The present study clearly demonstrates that drought stress at developmental stage of booting might be detrimental to the determination of sink size. As reported by Ludlow and Muchow (1990), plant responses to water stresses are influenced by the timing and intensity of the stress. As revealed by the tolerance/sensitivity scale and the yield-related traits measured here, the accessions we tested showed moderate to high sensitivity to drought stress. Because of the strong negative association of LDS with yield traits (illustrated in the path coefficient analysis; Fig. 3), we suggest that LDS can be used as a direct criterion in the selection of sorghum for drought tolerance. According to Cooper et al. (2006), a drought-tolerant genotype produces higher yields than a drought-susceptible genotype under water-stressed environments. The results here suggest that tolerant accessions may be utilized to develop elite breeding lines in sorghum breeding programs. DEH is a strong indication of sensitivity and is caused by growth retardation during soil drying in response to stress (Blum et al. 1999). Our results for DEF are supported by those of Sellamuthu et al. (2011), who reported that delay in flowering during the reproductive stage in rice could affect starch accumulation in grains by reducing photosynthesis and altering sink structure; these changes would reduce grain weight, and in turn, grain yield. The results of the present study suggest that non-zero values for DEH and DEF were due to reduction in photosynthesis because of high LDS, thus leading to reduction in grain yield. We suggest that DEH and DEF are among the three traits (LDS is the third) that most directly reflect phenotypic differences between individual accessions under control and drought stressed conditions. Among the genotypes tested, high variation was observed in POFE, showing that in sorghum, a water deficit at panicle initiation affects pollen development and can cause pollen sterility. The same result had been reported in barley (*Hordeum vulgare* L.), maize, wheat, and rice: when water potential was low during microsporogenesis and microgametogenesis, pollen grain sterility was observed (Saini 1997; Saini and Westgate 2000). However, drought-tolerant genotypes had higher values for POFE and related traits than did more sensitive genotypes (Table 2). Our POFE results were supported by Lafitte et al. (2003), who reported that spikelet fertility could be greatly affected when drought stress occurs near flowering. Based on the present results, POFE can be also used to assess drought tolerance among genotypes.

In this study we observed about 81 % decreases in yield GNOP under drought stress in comparison with the control conditions (Table 2). This decrease in yield showed high sensitivity of genotypes to water stress. In maize, moisture stress around anthesis could cause about a 22 % reduction in grain yield compared to the effects of moisture stress during the vegetative stage

(Denmead and Shaw 1960; McPherson and Boyer 1977; John 1990). Yield decreases were also reported in wheat and rice (Bingham 1966; Sheoran and Saini 1996). The results of the current study indicate a strong association between drought stress at developmental stage and performance.

QTLs under control conditions

LD-based genetic association studies offer a potential approach for mapping genes with modest effects (Hirschhorn and Daly 2005). Sorghum is primarily a self-pollinating crop and is expected to have high levels of LD and homozygosity, both of which facilitate LD mapping (Nordborg et al. 2002). Association mapping is a powerful method for identifying allelic variation because it explores the results of many generations of recombination and selection (Syvänen 2005). We found a number of QTLs by using four models of association mapping under both the control and drought stress conditions. Under the control condition, we found a total of 17 QTLs by both the Q and Q + K models on six chromosomes (chr-1, 2, 4, 8, 9 and 10) whereas no QTLs were identified on chr-3, 5, 6, and 7 (Fig. 4). Most of the QTLs were found on chr-1; in particular, QTLs for PH and TNOL were co-located at 68.7 cM, although no phenotypic correlation was observed between these traits (data not shown). The QTLs identified for PH in the present study are not supported by the previously reported QTL studies for this trait in sorghum but showed similarity with the results in rice (Sellamuthu et al. 2011). QTLs at identical positions were found for DTH and DTF on chr-1 at 107.1 cM, and a second QTL for DTH was found on chr-2. In another study of the same accessions, El Mannai et al. (2011) reported a QTL for DTF under natural conditions on chr-1 but at a different locus *Xtxp58* (68.7 cM) with $-\mathrm{Log}_{10}$ (P) value of 2.79 than the one identified in our study on chr-1 (107.1 cM) with

$-\mathrm{Log}_{10}$ (P) as 3.3. The detection of a QTL for the same trait on the same chromosome in multiple independent studies provides strong evidence for the validity of that QTL. We found two QTLs for LL on chr-1 and -4 and one QTL for LWI on chr-9. QTLs for LL and LWI on chr-4 and 6, respectively, were also identified by Feltus et al. (2006). Three QTLs for PL were found on chr-1, chr-2, and chr-10. Similarly, QTLs for PL were reported by other authors on chr-2 (Rami et al. 1998) and chr-1 (Shehzad et al. 2009b). These results showed that QTLs controlling PL are found at similar position in different studies. QTLs for two other traits, NOT and NOP, were found at the same locus on chr-2, consistent with the close association expected for these traits. Under the control conditions, the QTL for NOP on chr-2 explained 57.5 % of the phenotypic variation, with a $-\mathrm{Log}_{10}$ (P) value of 7.6. Hart et al. (2001) identified a QTL associated with the percentage of basal tillers with heads. The results of the present study suggested that the genetic basis of tillering was partly overlapping in diverse species. We also identified QTLs for POFE and 100GW/P on chr-4 and 8, respectively. The QTL for POFE was at the same position as one of the QTLs for LL, might suggest an association between these traits.

QTLs under drought stress conditions

Here, we identified several unique QTLs for various phenotypic traits under drought stress at developmental stage (Table 4; Fig. 4). Under the drought stress conditions, the Q and Q + K models identified QTLs on four chromosomes (chr-1, 2, 3 and 10); none were found on the remaining six chromosomes. Due to the low marker coverage for these chromosomes, there might be additional QTLs controlling traits under drought stress that could not be detected in the present study.

QTLs for traits TNOL, PL, NOP, and NOT were identified under both conditions, but QTLs for CD, LDS, PEX, and FLL were identified only under drought stress conditions; likewise, QTLs for DTH, DTF, LL, LW, PH and PWI identified only under control conditions. This indicates QTL × environment interactions and the complex genetic control of quantitative traits under stress conditions. PEX showed a strong correlation with GWP under drought stress, providing evidence for the importance of PEX in determining sorghum yield under stress. QTLs for PEX were not identified under control conditions, whereas two QTLs for PEX on chr-1 (59.0 cM) and chr-10 (55.5 cM) and one for CD on chr-1 (86.7 cM) were identified under the drought stress conditions. The QTL for PEX on chr-10 explained 61.2 % of the phenotypic variation for this trait. In rice, QTLs for PEX were identified under drought stress on chr-1 and 8 by Subashri et al. (2009) and on chr-1, 8, 10 and 11 by Yue et al. (2008).

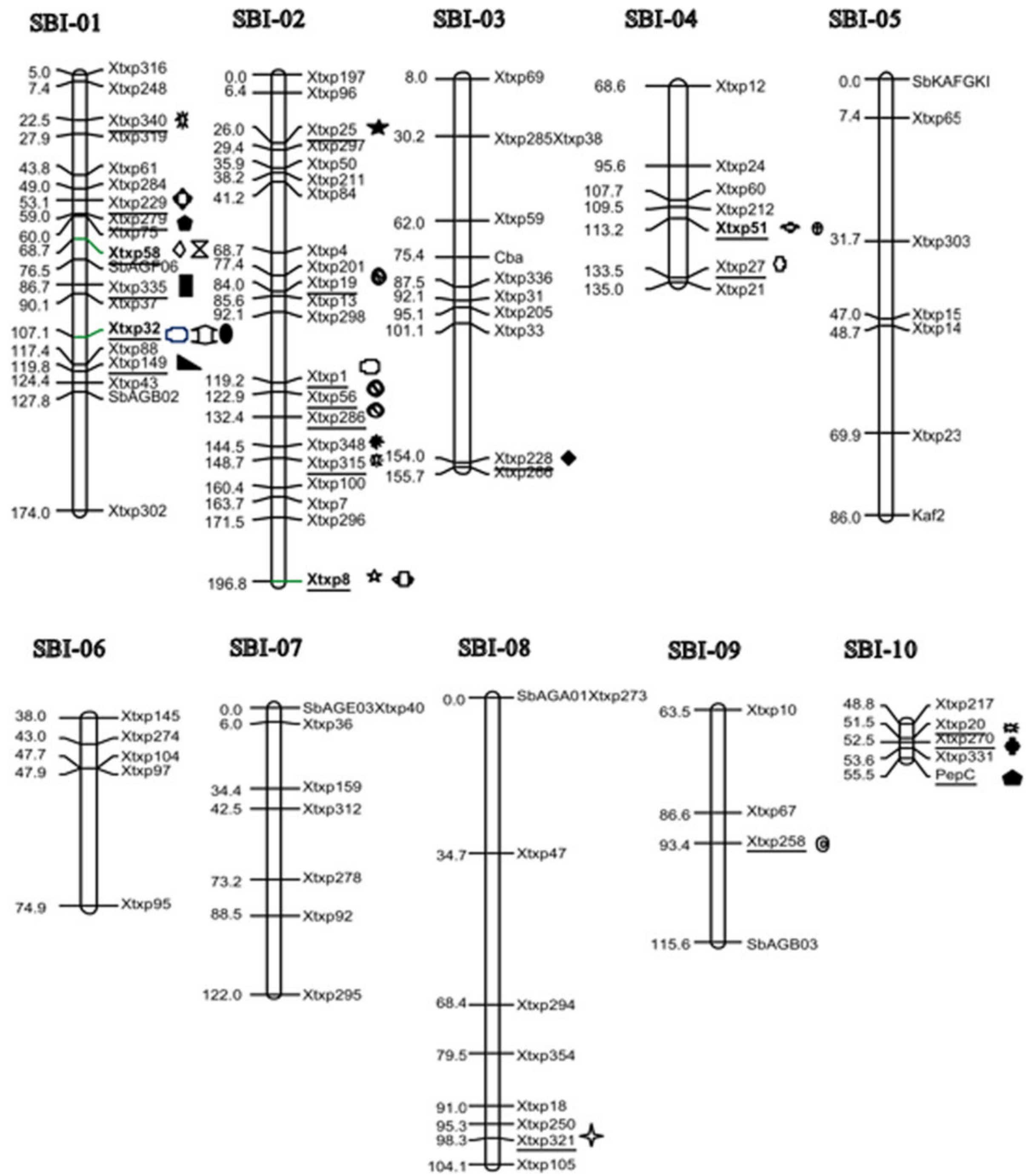


Fig. 4 Genetic linkage map of 98 sorghum SSR markers. *Empty symbols* indicate QTLs in control conditions; *filled symbols* indicate QTLs in drought stress conditions. PL, CD, DTF, DTH, NOP, PEX, LDS, 100GW, PWI, FLL, TNOL, PH, POFE, LL, NOT, LW

Under the drought stress conditions, no QTL for PH was identified even at a reduced threshold value of 2.2, despite the fact that a strong phenotypic correlation was observed between PH and various yield traits (Table 4; Fig. 3). Such differences in the QTL landscape may be due to the environmental sensitivity of certain genes. However, other factors such as sampling error due to population size, phenotypic evaluation, and errors in detection (e.g., the use of differing statistical thresholds to infer QTLs), can also contribute to the lack of congruency between studies, in terms of both estimated genomic location and the magnitude of genetic effects. For example, Feltus et al. (2006) detected a QTL for PEX on chr-3 under normal conditions, although here we detected a QTL for PEX only under the drought stress conditions. In present study some unique QTLs were found for FLL and LDS on chr-1 (107.1 and 119.8 cM, respectively) pertaining to a possible confounding effect of phenology-related QTLs. Under the drought stress conditions, no QTL had previously been reported in sorghum for LDS and FLL, whereas a QTL for FLL had been reported in drought stressed wheat on the short arm of chr-2D (Verma et al. 2004). In rice, QTLs for FLL were also identified on chr-2 and 3 under drought stress (Yue et al. 2008). Our results for LDS are in accordance with the detection of a QTL on rice chr-1 (Yue et al. 2005). QTLs for leaf rolling were identified in rice on chr-1, 3, 8 and 9 (Subashri et al. 2009). Hence, leaf drying and leaf rolling might play a similar role in maintaining internal plant water status.

Tuinstra et al. (1997) identified QTLs for stay-green on chr-1, 2 and 3 and for yield stability on chr-2 and 3 under the drought stress conditions, similar to our own observations on chr-1, 2 and 3 suggesting a positive relationship of LDS with stay-green and yield traits under drought stress conditions. A QTL for PL was found at the same position on chr-2 (148.7 cM) under both drought stress and control conditions. The persistence of this QTL in multiple environments and its association with yield indicates that this locus would be a valuable candidate for use in marker-assisted breeding. Other QTLs identified under the drought stress conditions were for NOT, TNOL, and NOP on chr-2, 3, and 10, respectively. Few of the QTLs could be physically mapped to the sorghum genome, and the biological relevance of those that could be mapped is unclear. The availability of additional sorghum genome sequence would make such analysis more powerful.

Homology with stay-green and other QTLs

We compared the published information on QTLs for stay-green and other yield-related traits identified in sorghum under drought stress with the QTLs identified in our study. We specifically focused on QTLs that were previously mapped on chr-1, 2, 3, and 10 under drought stress (ESM Table 2). This comparison revealed that a water deficit at reproductive stage has a direct association with stay-green trait which had been identified in sorghum under post-flowering drought stress. These results indicates some previously unreported genetic variations at these QTLs and its close association with the traits that we studied could be exploited by breeders to improve drought stress tolerance in sorghum.

Conclusions

In this study, we screened a sorghum diversity research set under drought stress conditions at developmental stage and identified QTLs for a variety of traits through association mapping. Grain yield under drought stress was influenced by many reproductive growth processes. Panicle- and pollen-related traits such as POFE, PEX, PL, and PWI had positive indirect effects on yield. These traits are candidates for selection in breeding for the improvement of drought resistance in sorghum. The QTLs we identified under water stress conditions were found in only four of the ten chromosomes (chr-1, 2, 3, and 10), suggesting that there are still gaps in the map due to the low marker coverage for some chromosomes. It is expected that additional studies using more markers will facilitate molecular breeding strategies for the improvement of drought tolerance in sorghum.

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